

Vigneshwara Chinnadurai

☎ 986-555-0901 | ✉ rockingstarvic@gmail.com | [in linkedin.com/in/vigneshwarac](https://www.linkedin.com/in/vigneshwarac) | github.com/VigneshwaraChinnadurai

Professional Summary

Senior AI Architect & Data Scientist with **8+ years** of experience designing enterprise-scale **Agentic AI, Generative AI**, and Machine Learning solutions. Pioneer in building **production-grade Autonomous AI Agent systems** using **AWS AgentCore SDK**, with expertise in **Hybrid RAG pipelines**, **LLM-powered self-critique loops**. Delivered **\$277M+** measurable business impact across Fortune 500 clients.

Experience

Senior AI Architect (AI and Data Science Engineer III)

Deloitte Consulting USI

January 2022 – Present

Bangalore, India

Disney PGT Document Parser Agent / AWS Lambda, Textract, Claude 3.7 Sonnet, SAM

- Architected **intelligent multi-format document parsing pipeline** for Disney's upfront spec sheet extraction, processing **Excel, PDF, Images, and Email (.msg)** with type-specific processors (pandas/openpyxl, Textract async/sync, extract-msg).
- Designed **serverless event-driven architecture**: S3 Upload → Lambda Router → File Processor → Textract/pandas → Smart Chunker → Claude (Bedrock) → DynamoDB, deployed via **AWS SAM/CloudFormation**.
- Implemented **smart chunking strategy** for large documents exceeding Claude's context limit with header context preservation and parallel multi-pass extraction with merged results.
- Engineered **composite Proposal ID generation** system linking disparate document types (specs, emails, PDFs) to unified proposal records using deterministic hashing when explicit IDs are absent.

Disney TVA Schema Intelligence Chatbot / AWS Bedrock, Claude 3.7 Sonnet, Agent Forge SDK

- Architected **production-grade conversational AI chatbot** using **custom S3VectorsAgentRuntime** extending AWS AgentCore SDK to handle S3 Vectors' unique **metadata.text** storage pattern that standard SDK couldn't parse.
- Implemented **intelligent token management** with configurable **max_rag_context_size** trimming that dynamically truncates retrieval chunks to prevent context overflow.
- Innovated **dual-path KB retrieval**: SDK's **extract_retrieval_chunks** for standard **content.text** + custom fallback for **metadata.text** with automatic deduplication.
- Created **follow-up query resolution system** with heuristic-based pronoun detection + URN extraction, rewriting vague queries for improved RAG retrieval.
- Reduced **memory retrieval latency by 70%** (17+ seconds to under 5 seconds) by optimizing memory hooks and implementing session stashing before RAG augmentation.
- Built **KB write-back pipeline** with deterministic vector keys enabling immediate queryability of agent-generated summaries without sync delays.

Disney ESPN Game Summary Platform / AWS Nova Premier, Lambda, Real-time Video AI

- Drove **\$231.76M AWS business expansion** by developing **GenAI highlight extraction platform**, architecting **7-step serverless Lambda pipeline**: MediaLive → Kinesis → MediaPackage → S3 → Nova Premier → DynamoDB → Highlights.
- Designed **multimodal prompt engineering** for video analysis with confidence scoring per field, play-type classification, and precise timestamp extraction. Built **clean_response_text()** handling markdown fences and partial JSON.
- Scaled platform to **4 major sports leagues** (NFL, Cricket, Rugby, Soccer), achieving **90% event capture rate**, by adapting multimodal prompt engineering for sport-specific play-type classification.

HP Enterprise - CXO Talk to Data Platform / Databricks Genie, Azure OpenAI GPT-4o-mini

- Eliminated **manual BI dashboard dependency** for C-suite decision-making by building **AI-powered conversational analytics platform** with first implementation of **Databricks Genie's NL-to-SQL** capabilities and user token forwarding.
- Reduced **LLM hallucination rate** through output validation against schema definitions, by architecting **multi-table semantic layer** (2 Fact + 11 Dimension tables) with YAML-based semantic definitions.

Toyota Demand Forecasting Engine / LSTM, N-BEATS, Prophet, PySpark

- Reduced **annual inventory costs by \$46.1 Million** with 70% WMAPE across 15 distribution centers, by engineering enterprise-scale forecasting system managing **1.2 million SKU combinations**.
- Implemented **hybrid LSTM architecture** with dual-layer Lambda expansions, **zero-entry "+1 trick"** for MinMax scaling, and **ensemble model selection** (LSTM/N-BEATS/SVR/Prophet) per SKU characteristics.
- Accelerated **forecast execution throughput** across multiple environments by designing **orchestrator.py** with **multiprocessing pool** and YAML-driven configuration.

Halliburton Financial Predictive Modeling / N-BEATS, SARIMAX, Deep Learning

- Achieved **98% accuracy** in invoice dispute prediction by developing **deep learning forecasting solution** for Accounts Payable/Receivable using **N-BEATS and SARIMAX** ensemble.
- Reduced financial forecast **delivery time by 98%** by building automated pipeline with multi-variate time series modeling capturing both trend and seasonality patterns.

Morningstar PDF Layout Intelligence / Document Image Transformer (DIT), Computer Vision

- Accelerated **PDF title extraction accuracy** for downstream business workflows by implementing **Document Image Transformer (DIT)** for automated structural layout analysis across diverse document formats.
- Eliminated **manual document review bottleneck**, enabling business teams to consume structured title metadata, by building a **high-speed layout detection engine** that correctly identifies and classifies document titles from raw PDF inputs.

Firm Initiatives: Agentic AI & MLOps Estimation Frameworks

- Standardized **Agentic AI and MLOps project scoping** across multiple Deloitte practices, validated through **24+ iterations** of refinement and firm-wide adoption, by creating **industry-first estimation frameworks** for consistent project sizing.

Firm Initiatives: UBIX AI Platform Evaluation

- Evaluated **UBIX AI** platform's strategic fit within Data Science practice, as validated by **comprehensive benchmarking against traditional manual algorithms**, by conducting rigorous performance testing to provide data-driven insights enabling SAP Practice Leader to finalize a **strategic enterprise AI partnership**.

Senior Software Engineer (Machine Learning)

WIPRO Technologies

February 2018 – January 2022

Bangalore, India

WIPRO Holmes (R&D) / LSTM, Reg-exp, Deep Learning

- Improved **legal clause extraction accuracy from 76% to 93%** in Contract Review tool by enhancing with **Bi-LSTM + CRF architecture** for sequence labeling in MSAs and SOWs. The **bidirectional LSTM** captured long-range contextual dependencies while the **CRF layer** learned valid BIO tag transitions (B-CLAUSE → I-CLAUSE), preventing invalid sequence predictions. Combined with **character-level embeddings** for OOV legal terminology and **domain-specific regex patterns** for structured entities (dates, parties, amounts).

Huawei Video (R&D) / DIP, Vector DB, Deep Learning

- Powered **3 real-time recommendation pipelines (Hot Broadcast, You May Like, Waterfall E2)** integrated with **RASE (Huawei Recommendation-as-a-Service)**, by creating data model using **Deep Interest Network (DIN)** with **attention-weighted embeddings** capturing user-item interactions, stored in **Redis** for low-latency serving + **Milvus** for vector similarity retrieval.

Shell Plc / CNN, Transfer Learning, Machine Learning

- Automated **2D-to-3D schematic conversion** for P&ID drawings by building **CNN with VGG19 Transfer Learning** in Keras to detect small symbols (valves, pumps, instruments), combined with **Hough Circle Transform + TensorFlow Object Detection API** for geometric shape recognition.
- Digitized **legacy engineering documents** by developing **OCR pipeline** combining **Tesseract** (for structured text) + **AWS Textract** (for tables/forms) with custom image preprocessing (contrast enhancement, adaptive thresholding, noise reduction).

Personal R&D Projects

Autonomous Multi-Agent Strategic Analysis Engine | Gemini 2.5 Flash, Hybrid RAG, Self-Refinement

- **Genuinely Agentic Architecture:** LLM-powered planning where **AnalystAgent** autonomously decomposes complex prompts into executable multi-step plans without predefined workflows.
- **Hybrid Retrieval Pipeline:** Custom BM25 sparse + Gemini dense embeddings with **Reciprocal Rank Fusion (RRF)** + **LLM-powered reranking** (0-1 relevance scoring).
- **Self-Critique & Refinement Loop:** Rubric-based critique (Depth/Evidence/Causality/Actionability) with automatic **NEEDS_REFINEMENT** detection triggering re-retrieval cycles.
- **Dynamic Tool Generation:** Runtime code synthesis creating callable Python functions from LLM output. **Graph Memory** with semantic triple extraction for persistent knowledge.
- **Real data integration:** SEC EDGAR XBRL, Google News RSS, Gemini Search Grounding. **Cost Guard System** with budget caps and aggressive caching.

- Quantum Portfolio Optimization Engine** (MBA Project) | *D-Wave Quantum Annealing, QUBO, Simulated Annealing*
- **QUBO Formulation from First Principles:** Quadratic Unconstrained Binary Optimization with risk terms (covariance), return rewards (mean returns), and cardinality constraints via penalty functions.
 - **Classical-Quantum Benchmark:** Side-by-side scipy max-Sharpe vs. quantum annealing with hyperparameter sweep over *risk & return*, penalty coefficients.
 - **D-Wave Leap QPU Integration:** Production-ready execution with **EmbeddingComposite** for automatic qubit mapping. Efficient frontier visualization.

- LeetCode AI Companion (Android) (Personal Development)** | *Kotlin, Jetpack Compose, Gemini API, GraphQL*
- **Jetpack Compose UI** with Material Design 3. LeetCode GraphQL integration for daily challenges. **Gemini generation pipeline** with retry logic.
 - **GitHub Push Automation:** One-tap export of solutions to repository. Calendar event integration for completion tracking. Mermaid flow viewer with gestures.

Education

Manipal University <i>Master of Business Administration, Analytics and Data Science</i>	Jaipur, India <i>August 2024 – June 2026</i>
Anna University <i>Bachelor of Engineering, Electrical & Electronics</i>	Chennai, India <i>August 2014 – May 2017</i>

Certifications

- AWS Certified Machine Learning – Specialty** | Amazon Web Services (Oct 2021)
Delivery Accreditation - AI Agents | ServiceNow
Configure AI Applications to Optimize Search Results | Google
Deep Learning Specialization | Coursera/Udacity

Technical Skills

- Agentic AI & RAG:** Agent Forge SDK (AWS AgentCore), Strands Agents, Multi-Agent Orchestration, LLM Planning & Critique, Self-Refinement Loops, Hybrid Retrieval (BM25 + Dense + RRF), LLM Reranking, S3 Vectors, Token-aware Trimming
- LLMs & Foundation Models:** Claude 3.7 Sonnet, GPT-4/4o-mini, AWS Nova Premier, Gemini 2.5 Flash, Llama 3, Amazon Titan Embed v2
- Quantum Computing:** D-Wave Quantum Annealing, QUBO Formulation, Simulated Annealing, D-Wave Leap, Binary Quadratic Models
- Deep Learning:** Transformers (BERT, DiT), CNN (VGG19, ResNet), LSTM, Bi-LSTM, Deep Interest Network (DIN), N-BEATS, Prophet
- Cloud & MLOps:** AWS (Bedrock, SageMaker, Lambda, S3, MediaLive, Kinesis, SQS, Identity Manager, Media Package, Media Convert), Azure (OpenAI), GCP (Vertex AI), Docker, Kubernetes, Airflow, DataBricks
- Languages & Data:** Python, PySpark, SQL, Kotlin (Android), Pandas, NumPy, TensorFlow, PyTorch

Key Achievements

\$46.1M annual inventory savings at Toyota | **\$231.76M** AWS expansion via Disney ESPN | **30%** latency reduction (70s→50s) | **93%** contract extraction accuracy | **1.2M SKUs** forecasting scale | **4+ Fortune 500** production deployments